

PURE MATHEMATICS WITH MECHANICS • UPPER SIXTH

UPPER SIXTH — PURE MATHEMATICS WITH MECHANICS

CHAPTER 5

FIRST ORDER DIFFERENTIAL EQUATIONS WITH SEPARABLE VARIABLES

SUGGESTED TIME: 6 PERIODS

IN THIS CHAPTER

This chapter introduces first-order differential equations with separable variables, guiding students from the basic definition through recognition, separation, integration, and the use of initial conditions to obtain particular solutions. It concludes with concrete applications drawn from physics, biology and everyday Cameroonian contexts.

- Define a first-order differential equation and distinguish between general and particular solutions.
- Recognise whether a given first-order equation is separable.
- Apply the separation-of-variables technique correctly, including handling constant solutions.
- Integrate both sides of a separated equation using appropriate integration methods.
- Express the general solution in implicit or explicit form and verify it by differentiation.
- Use an initial condition to determine the arbitrary constant and write the particular solution.



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CHAPTER OBJECTIVES

- Define a first-order differential equation and distinguish between general and particular solutions.
- Recognise whether a given first-order equation is separable.
- Apply the separation-of-variables technique correctly, including handling constant solutions.
- Integrate both sides of a separated equation using appropriate integration methods.
- Express the general solution in implicit or explicit form and verify it by differentiation.
- Use an initial condition to determine the arbitrary constant and write the particular solution.
- Model real-world phenomena (cooling, population growth, mixing) with separable differential equations.
- Interpret the mathematical solution in the context of the original problem and discuss its domain of validity.

PREREQUISITES

- Differentiation rules (power, exponential, trigonometric, chain rule).
- Integration techniques: substitution, partial fractions, standard integrals.
- Algebraic manipulation of equations and handling of absolute values.

- Basic understanding of functions, graphs, and slope fields.
- Familiarity with the concept of a constant of integration.

1 MEANING OF FIRST ORDER DIFFERENTIAL EQUATIONS

Imagine a steaming pot of ndolé left on the kitchen table in Yaoundé. As it cools, its temperature drops faster at first and then more slowly, following a natural law that can be described by a first-order differential equation. In the same way, the spread of a rumour in a village or the growth of a cassava crop can be modelled with equations where the rate of change depends on the current state. This chapter equips you to translate such real-life situations into mathematics and to solve them.

1.1 What is a Differential Equation?

A **differential equation** is an equation that relates an unknown function to one or more of its derivatives. The function usually represents a quantity that varies (temperature, population, velocity, ...) and the derivatives represent its rates of change.

DEFINITION Differential Equation

A differential equation is an equation involving an unknown function $y = y(x)$ and its derivatives $\frac{dy}{dx}, \frac{d^2y}{dx^2}, \dots$. The order of the differential equation is the order of the highest derivative that appears. The degree is the power of the highest derivative after the equation has been made rational and polynomial in all derivatives.

For example:

$$\frac{dy}{dx} = 3x^2 \quad (\text{order 1, degree 1})$$

$$\left(\frac{d^2y}{dx^2}\right)^2 + y = \sin x \quad (\text{order 2, degree 2})$$

1.2 Order and Degree

The **order** tells us how many times we must differentiate the unknown function to obtain the equation. The **degree** is only defined when the equation is a polynomial in the derivatives; it is the exponent of the highest-order derivative.

WARNING Order vs. Degree

A very common mistake is to confuse the order with the degree. Remember:

- Order = highest derivative present (first, second, ...).
- Degree = power of that highest derivative after clearing fractions and radicals.

For instance, $\left(\frac{dy}{dx}\right)^3 + y = x$ has order 1 and degree 3, whereas $\frac{d^2y}{dx^2} + \sqrt{\frac{dy}{dx}} = 0$ has order 2 but its degree is not defined (because of the square root).

1.3 First-Order Differential Equations

A **first-order differential equation** involves only the first derivative $\frac{dy}{dx}$. It can always be written in the form

$$\frac{dy}{dx} = f(x, y)$$

where f is a given function of the independent variable x and the dependent variable y . The variable x is the **independent variable** (often time or position) and y is the **dependent variable** (the quantity we want to find).

Simple examples:

- $\frac{dy}{dx} = 3x^2$ — the rate of change depends only on x .
- $\frac{dy}{dx} = y$ — the rate of change is proportional to the current value (exponential growth/decay).
- $\frac{dy}{dx} = x + y$ — the rate of change depends on both x and y .

1.4 General and Particular Solutions

When we solve a first-order differential equation, we obtain a **general solution** — a family of functions containing one arbitrary constant (usually denoted C). Each choice of C gives a **particular solution**, which is a single curve in the xy -plane.

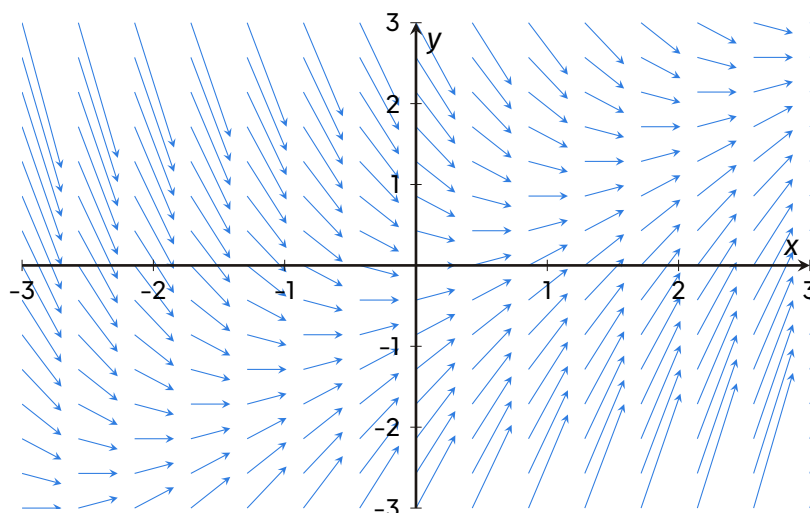
Geometrically, the differential equation $\frac{dy}{dx} = f(x, y)$ assigns a slope $f(x, y)$ to every point (x, y) in the plane. The general solution is the family of all curves whose tangent at each point has exactly that slope. A particular solution is the unique curve from that family that passes through a given point (x_0, y_0) — this point is called an **initial condition**.

1.5 Slope Fields (Direction Fields)

A **slope field** (or **direction field**) is a graphical tool that helps us visualise the behaviour of solutions without solving the equation analytically. At a grid of points (x_i, y_j) we draw a short line segment whose slope equals $f(x_i, y_j)$. The collection of these segments shows the “flow” of the solutions.

METHOD Constructing a Slope Field

1. Choose a rectangular grid of points (x_i, y_j) in the region of interest.
2. At each grid point compute the slope $m_{ij} = f(x_i, y_j)$.
3. Draw a short line segment centred at (x_i, y_j) with slope m_{ij} (length can be constant or scaled).
4. Optionally, sketch a few integral curves that follow the direction of the segments.



Slope field for $\frac{dy}{dx} = x - y$. The short arrows indicate the direction of the solution curves at each point.

The slope field above for $\frac{dy}{dx} = x - y$ shows that solutions tend to approach the line $y = x$ as x increases. This qualitative insight is often valuable before attempting an analytic solution.

1.6 Integral Curves

The curves that are everywhere tangent to the slope field are called **integral curves**. For the simple equation $\frac{dy}{dx} = y$ the general solution is $y = Ce^x$. Plotting several members of this family on the same axes illustrates how the arbitrary constant C shifts the curve vertically.



Integral curves of $\frac{dy}{dx} = y$ for several values of the constant C . All curves have the same exponential shape, differing only by a vertical scaling factor.

1.7 Initial Conditions and Particular Solutions

An **initial condition** specifies the value of the unknown function at a particular point: $y(x_0) = y_0$. Substituting this condition into the general solution determines the constant C uniquely, yielding a particular solution.

PROPERTY Existence and Uniqueness (Brief Statement)

If the function $f(x, y)$ and its partial derivative $\frac{\partial f}{\partial y}$ are continuous in a rectangle containing the point (x_0, y_0) , then the initial value problem

$$\frac{dy}{dx} = f(x, y), \quad y(x_0) = y_0$$

has a unique solution defined on some interval around x_0 .

This theorem guarantees that, under mild conditions, the physical situations we model (cooling ndolé, spreading rumour, growing cassava) have a well-defined future evolution once the present state is known.

1.8 Worked Example

WORKED EXAMPLE 1 Finding a Particular Solution

Statement: Solve the initial value problem

$$\frac{dy}{dx} = 3x^2, \quad y(1) = 4.$$

SOLUTION

Detailed solution:

1. Recognise the equation. It is a first-order differential equation where the right-hand side depends only on x .
2. Integrate both sides with respect to x .

$$y = \int 3x^2 dx = x^3 + C,$$

where C is an arbitrary constant. This is the general solution.

3. Apply the initial condition $y(1) = 4$.

$$4 = 1^3 + C \implies C = 3.$$

4. Write the particular solution.

$$y = x^3 + 3.$$

5. Check. $\frac{dy}{dx} = 3x^2$ and $y(1) = 1 + 3 = 4$. Both the differential equation and the initial condition are satisfied.

1.9 Common Mistakes

WARNING Frequent Errors

- Forgetting the constant of integration. Every indefinite integration introduces an arbitrary constant; omitting it loses the general solution.
- Confusing the independent and dependent variables. In $\frac{dy}{dx} = f(x, y)$, x is independent, y is dependent. Swapping them changes the problem entirely.
- Misinterpreting the slope field. The short segments represent the direction of the solution curve at that point, not the curve itself. Do not join the tips of the segments; instead, sketch curves that are everywhere tangent to them.
- Assuming a unique solution without checking continuity. The existence–uniqueness theorem requires f and $\frac{\partial f}{\partial y}$ to be continuous near the initial point. If they are not, multiple solutions or no solution may exist.

1.10 Summary of Key Points

KEY POINTS Key Points for This Section

- A differential equation relates an unknown function to its derivatives.
- The order is the highest derivative; the degree is the power of that derivative (when defined).
- A first-order equation has the form $\frac{dy}{dx} = f(x, y)$.
- The general solution contains one arbitrary constant and represents a family of curves (integral curves).
- A particular solution is obtained by imposing an initial condition $y(x_0) = y_0$.
- Slope fields give a visual picture of the direction of solutions at each point.
- Under continuity of f and $\partial f / \partial y$, the initial value problem has a unique solution.

2 RECOGNITION OF SEPARABLE EQUATIONS

In the previous section we learned what a first-order differential equation is and how it arises from modelling rates of change. We now turn to a particularly friendly family of such equations: those whose variables can be **separated**. Recognising this structure is the first decisive step towards an explicit solution.

2.1 What Makes a Differential Equation Separable?

DEFINITION Separable First-Order Differential Equation

A first-order differential equation is called separable if it can be written in the form

$$\frac{dy}{dx} = g(x) h(y)$$

or, equivalently,

$$M(x) N(y) dx + P(x) Q(y) dy = 0,$$

where the functions of x and the functions of y appear only as products. After a simple algebraic rearrangement we obtain

$$\frac{1}{h(y)} dy = g(x) dx \quad \text{or} \quad f(x) dx = g(y) dy.$$

The key idea: all terms containing x (and dx) can be gathered on one side, and all terms containing y (and dy) on the other. Once this is achieved, each side can be integrated independently.

2.2 Algebraic Test for Separability

Given a differential equation, follow these algebraic steps to decide whether it is separable:

1. Write the equation explicitly as $\frac{dy}{dx} = F(x, y)$.
2. Attempt to factor $F(x, y)$ into a product $g(x) h(y)$. This may require elementary algebraic manipulations: expanding brackets, cancelling common factors, or rewriting a quotient as a product.
3. If such a factorisation exists, the equation is separable. Divide both sides by $h(y)$ (provided $h(y) \neq 0$) and multiply by dx to obtain the separated form

$$\frac{1}{h(y)} dy = g(x) dx.$$

4. If no factorisation into a pure product of a function of x and a function of y is possible, the equation is not separable (though it might become separable after a suitable substitution — see the tip at the end of this section).

2.3 Examples of Separable Equations

EXAMPLE Separable: $dy/dx = xy$

$$\frac{dy}{dx} = xy \implies \frac{1}{y} dy = x dx.$$

Here $g(x) = x$ and $h(y) = y$. The variables are already separated after dividing by y .

EXAMPLE Separable: $dy/dx = \frac{\sin x}{y^2}$

$$\frac{dy}{dx} = \frac{\sin x}{y^2} \implies y^2 dy = \sin x dx.$$

We have $g(x) = \sin x$ and $h(y) = 1/y^2$. Multiplying by $y^2 dx$ yields the separated form.

EXAMPLE Separable after simplification: $\frac{dy}{dx} = \frac{x^2y + y}{x}$

$$\frac{dy}{dx} = \frac{y(x^2 + 1)}{x} = \underbrace{\frac{x^2 + 1}{x}}_{g(x)} \underbrace{y}_{h(y)}.$$

Factorising the numerator reveals the product structure.

2.4 Examples of Non-Separable Equations

EXAMPLE Not separable: $dy/dx = x + y$

The right-hand side is a sum of a function of x and a function of y . It cannot be expressed as a product $g(x)h(y)$ (except in trivial cases where one factor is constant, which would make the equation either purely x -dependent or purely y -dependent).

EXAMPLE Not separable: $dy/dx = \frac{xy}{x + y}$

The denominator $x + y$ mixes the variables in a way that prevents factorisation into a product of a function of x alone and a function of y alone.

2.5 Hidden Separability — Simple Algebraic Manipulation

Sometimes an equation does not look separable at first glance, but a little algebra reveals the structure.

EXAMPLE Hidden separability: $\frac{dy}{dx} = \frac{y^2 - 1}{x^2 - 1}$

$$\frac{dy}{dx} = \frac{y^2 - 1}{x^2 - 1} = \frac{1}{\underbrace{x^2 - 1}_{g(x)}} \underbrace{(y^2 - 1)}_{h(y)}.$$

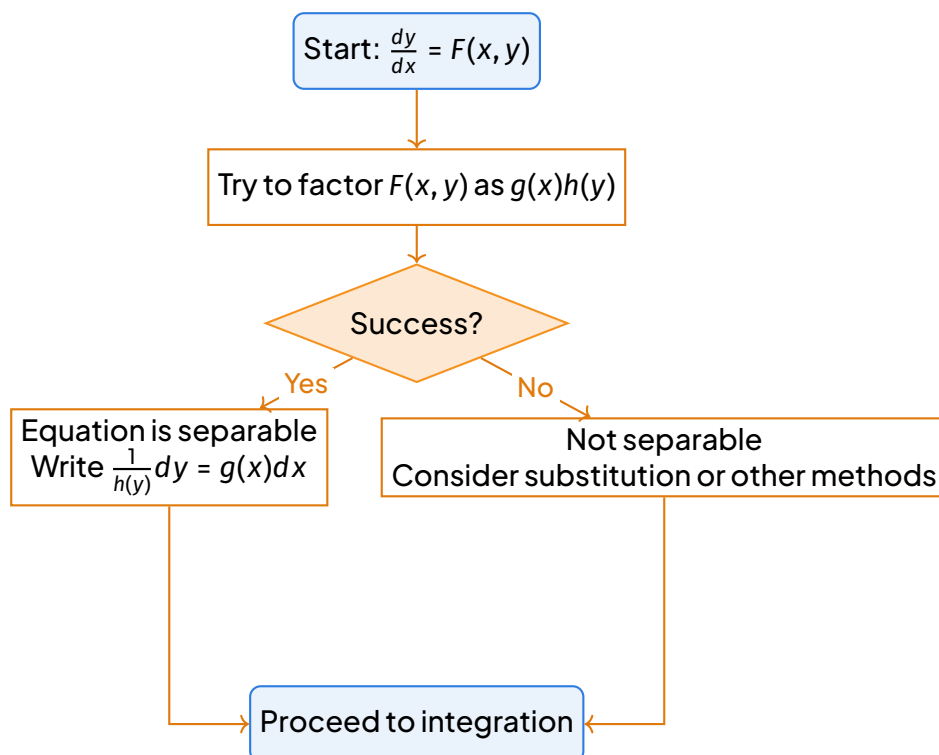
The quotient of two differences of squares is already a product of a function of x and a function of y .

EXAMPLE Hidden separability: $\frac{dy}{dx} = \frac{e^{x+y}}{x}$

$$\frac{dy}{dx} = \frac{e^x e^y}{x} = \frac{e^x}{\underbrace{x}_{g(x)}} \underbrace{e^y}_{h(y)}.$$

Using the law of exponents $e^{x+y} = e^x e^y$ exposes the separable form.

2.6 Flowchart for Testing Separability



Flowchart: steps to test whether a first-order DE is separable.

2.7 Method: Step-by-Step Recognition

METHOD Recognising a Separable Differential Equation

1. Write the DE in the explicit form $\frac{dy}{dx} = F(x, y)$.
2. Inspect $F(x, y)$: can it be written as a product $g(x)h(y)$? Use algebraic identities (factorisation, exponent laws, trigonometric identities) if necessary.
3. If yes, the DE is separable. Divide both sides by $h(y)$ (noting any values where $h(y) = 0$) and multiply by dx to obtain $\frac{1}{h(y)} dy = g(x) dx$.
4. If no, the DE is not separable. Record this and consider whether a substitution (e.g. $v = y/x$ for homogeneous equations) might render it separable.

2.8 Important Warning: Lost Constant Solutions

WARNING Dividing by $h(y)$ May Lose Solutions

When we divide by $h(y)$ we implicitly assume $h(y) \neq 0$. If $h(y_0) = 0$ for some constant y_0 , then $y(x) \equiv y_0$ is a constant solution of the original DE (since $dy/dx = 0$ and the right-hand side becomes $g(x) \cdot 0 = 0$). These constant solutions are not captured by the separated integration and must be checked separately.

2.9 Key Points to Remember**KEY POINTS** Separable Equations — Essentials

- A first-order DE is separable iff it can be written as $\frac{dy}{dx} = g(x)h(y)$.
- The separated form is $\frac{1}{h(y)} dy = g(x) dx$ (or $f(x) dx = g(y) dy$).
- Always check for constant solutions $y = y_0$ where $h(y_0) = 0$ before dividing by $h(y)$.
- Common algebraic tricks: factorisation, $e^{x+y} = e^x e^y$, $\ln(xy) = \ln x + \ln y$, trigonometric identities.
- Sums like $x+y$ or mixed terms like x^2+y^2 in denominators generally prevent separability.

2.10 Worked Example: Recognising and Solving a Separable DE**WORKED EXAMPLE 2** Recognise and solve $\frac{dy}{dx} = \frac{2x}{y(1+x^2)}$

Statement. Determine whether the differential equation

$$\frac{dy}{dx} = \frac{2x}{y(1+x^2)}$$

is separable. If it is, find its general solution and identify any constant solutions.

Detailed solution.

1. Write in explicit form. The equation is already given as $dy/dx = F(x, y)$ with

$$F(x, y) = \frac{2x}{y(1+x^2)}.$$

2. Factor $F(x, y)$ as $g(x)h(y)$. Observe that

$$\frac{2x}{y(1+x^2)} = \underbrace{\frac{2x}{1+x^2}}_{g(x)} \underbrace{\frac{1}{y}}_{h(y)}.$$

Thus the DE is separable with $g(x) = \frac{2x}{1+x^2}$ and $h(y) = \frac{1}{y}$.

3. Check for constant solutions. $h(y) = 1/y = 0$ has no real solution, so no constant solutions are lost by division.
4. Separate the variables.

$$\frac{1}{h(y)} dy = g(x) dx \implies y dy = \frac{2x}{1+x^2} dx.$$

5. Integrate both sides.

$$\int y dy = \int \frac{2x}{1+x^2} dx.$$

Left side: $\frac{y^2}{2}$. Right side: substitute $u = 1 + x^2$, $du = 2x dx$ giving $\int \frac{du}{u} = \ln |u| = \ln(1 + x^2)$ (since $1 + x^2 > 0$). Hence

$$\frac{y^2}{2} = \ln(1 + x^2) + C,$$

where C is an arbitrary constant.

6. Express the general solution.

$$y^2 = 2 \ln(1 + x^2) + 2C \implies y = \pm \sqrt{2 \ln(1 + x^2) + C'},$$

with $C' = 2C$ an arbitrary constant. The $\pm$ reflects the two branches obtained when taking the square root. The solution is defined for $2 \ln(1 + x^2) + C' \geq 0$.

7. Final answer. The DE is separable. Its general solution is

$$y = \pm \sqrt{2 \ln(1 + x^2) + C}, \quad C \in \mathbb{R},$$

and there are no constant solutions.

2.11 Practice Set: Decide Separability

EXERCISE 1 Separable or not?

For each of the following differential equations, decide whether it is separable. If it is, write it in the separated form $f(x) dx = g(y) dy$. If it is not, give a brief reason.

1. $\frac{dy}{dx} = x^2 y^3$
2. $\frac{dy}{dx} = \sin(x + y)$

3. $\frac{dy}{dx} = \frac{e^x}{y^2 + 1}$
4. $\frac{dy}{dx} = \frac{x}{y} + \frac{y}{x}$
5. $\frac{dy}{dx} = \frac{xy}{x^2 + y^2}$
6. $\frac{dy}{dx} = \frac{\cos x}{\sin y}$
7. $\frac{dy}{dx} = \frac{y^2 - 4}{x^2 - 9}$
8. $\frac{dy}{dx} = \frac{x + y}{x - y}$
9. $\frac{dy}{dx} = \frac{e^{x-y}}{x}$
10. $\frac{dy}{dx} = \frac{1}{x} + \frac{1}{y}$

2.12 Answers and Brief Comments

SOLUTION Practice Set

1. Separable. $g(x) = x^2$, $h(y) = y^3$. Separated: $\frac{1}{y^3} dy = x^2 dx$.
2. Not separable. $\sin(x + y)$ cannot be factorised into a product of a function of x and a function of y .
3. Separable. $g(x) = e^x$, $h(y) = \frac{1}{y^2 + 1}$. Separated: $(y^2 + 1) dy = e^x dx$.
4. Not separable. The sum $\frac{x}{y} + \frac{y}{x} = \frac{x^2 + y^2}{xy}$ mixes variables; no factorisation into $g(x)h(y)$.
5. Not separable. Denominator $x^2 + y^2$ prevents separation.
6. Separable. $g(x) = \cos x$, $h(y) = \frac{1}{\sin y}$. Separated: $\sin y dy = \cos x dx$.
7. Separable. $g(x) = \frac{1}{x^2 - 9}$, $h(y) = y^2 - 4$. Separated: $\frac{1}{y^2 - 4} dy = \frac{1}{x^2 - 9} dx$.
8. Not separable. The quotient $\frac{x + y}{x - y}$ is not a product of separate functions.

9. Separable. $e^{x-y} = e^x e^{-y}$. $g(x) = \frac{e^x}{x}$, $h(y) = e^{-y}$. Separated: $e^y dy = \frac{e^x}{x} dx$.

10. Not separable. Sum of a function of x and a function of y cannot be written as a product.

2.13 Looking Ahead

Now that we can confidently recognise separable equations, the next section will show how to carry out the integration step to obtain the **general solution**, and how to use initial conditions to pick out a **particular solution**. We will also explore a variety of real-world applications — from population growth in the Adamawa region to cooling of a cup of Cameroonian coffee — where separable differential equations provide exact mathematical models.

3 SEPARATION OF VARIABLES

This section covers **Lessons 63 and 64** of the official syllabus: the technique of separating variables and integrating to obtain the general solution of a first order separable differential equation. In the previous section we learned to recognise a separable equation: one that can be written in the form $\frac{dy}{dx} = g(x)h(y)$, where the right-hand side splits into a factor depending only on x and a factor depending only on y . Recognition, however, is only the first step. We now learn the central skill of this chapter: how to solve such an equation by **separating the variables** — gathering everything involving y with dy on one side, and everything involving x with dx on the other — and then integrating both sides. This single idea, simple as it looks, unlocks an enormous family of differential equations, from population growth to radioactive decay and Newton's law of cooling.

3.1 The Formal Separation Procedure

DEFINITION Separation of variables

Let

$$\frac{dy}{dx} = g(x)h(y)$$

be a first order separable differential equation, where g is continuous on an interval I and h is continuous on an interval J . On any interval where $h(y) \neq 0$, we may divide both sides by $h(y)$ and write the equation in separated form:

$$\frac{1}{h(y)} dy = g(x) dx.$$

Integrating both sides with respect to their own variable gives

$$\int \frac{dy}{h(y)} = \int g(x) dx + C,$$

that is, an implicit relation $F(y) = G(x) + C$ between x and y , where $F'(y) = 1/h(y)$ and $G'(x) = g(x)$.

Why is this legitimate?

A careful justification uses the chain rule. Suppose F is an antiderivative of $\frac{1}{h}$, so $F'(y) = \frac{1}{h(y)}$. If y is a function of x satisfying $\frac{dy}{dx} = g(x)h(y)$, then

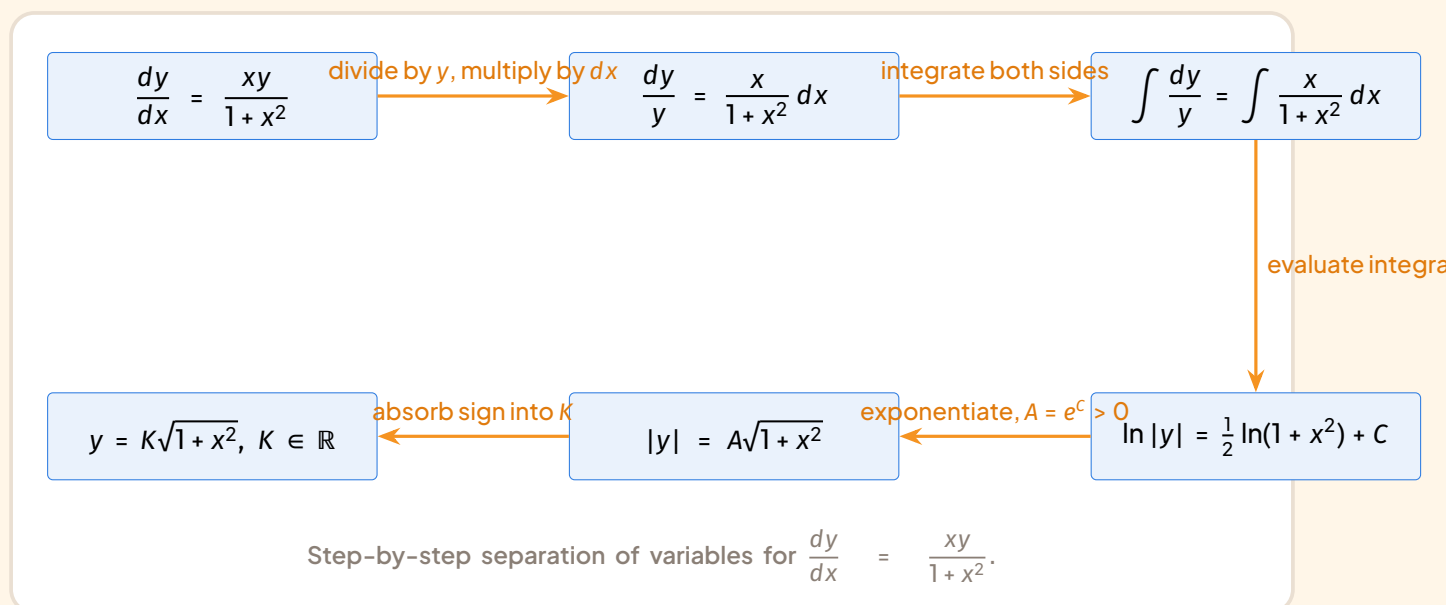
$$\frac{d}{dx}[F(y)] = F'(y) \frac{dy}{dx} = \frac{1}{h(y)} \cdot g(x)h(y) = g(x).$$

Hence $F(y)$ is an antiderivative of $g(x)$, so $F(y) = G(x) + C$ where $G'(x) = g(x)$. This is exactly what the symbolic manipulation “ $\frac{dy}{h(y)} = g(x) dx$, then integrate” produces. The Leibniz notation $\frac{dy}{dx}$ is doing real work for us here: treating dy and dx as separable quantities is a reliable shortcut, and the chain rule guarantees that the shortcut is sound.

METHOD Separation of variables — the standard routine

To solve $\frac{dy}{dx} = g(x)h(y)$:

1. Write the equation in separable form. Identify $g(x)$ and $h(y)$ explicitly.
2. Isolate the differentials. Divide by $h(y)$ (noting the values of y for which $h(y) = 0$) and multiply by dx to obtain $\frac{dy}{h(y)} = g(x) dx$.
3. Integrate both sides. Compute $\int \frac{dy}{h(y)}$ and $\int g(x) dx$.
4. Add one constant of integration. A single constant C on one side is enough, since two constants C_1, C_2 merge into $C = C_2 - C_1$.
5. Make y the subject if possible, and state the constant solutions (if any) found in step 1.



3.2 Do Not Forget the Constant (Equilibrium) Solutions

When we divided by $h(y)$ in the very first step, we silently assumed $h(y) \neq 0$. But what happens at the values of y where $h(y) = 0$? If $h(y_0) = 0$ for some constant y_0 , then the constant function $y(x) = y_0$ has derivative $\frac{dy}{dx} = 0$, and the right-hand side is $g(x)h(y_0) = g(x) \cdot 0 = 0$. So the equation is satisfied identically.

PROPERTY Constant solutions

If $h(y_0) = 0$, then the constant function $y = y_0$ is a solution of $\frac{dy}{dx} = g(x)h(y)$ on the whole interval where g is defined. These are called **constant solutions** or **equilibrium solutions**. (The verification is examinable: substitute and check both sides vanish.)

WARNING The incomplete general solution

Dividing by $h(y)$ without checking where $h(y) = 0$ is the most common way to lose solutions. For example, in $\frac{dy}{dx} = y^2 \sin x$, dividing by y^2 throws away the solution $y = 0$, which is perfectly valid. Always ask: "Can $h(y)$ be zero? If so, is that constant a solution?" Some of these constant solutions are recovered from the general formula for a special value of the constant (e.g. $K = 0$ in $y = K\sqrt{1+x^2}$); others, like $y = 0$ for $\frac{dy}{dx} = y^2 \sin x$, are singular solutions that no choice of C will ever produce.

3.3 Integration: From Separated Form to Implicit Relation

After separation, each side is integrated with respect to its own variable. The result is a relation

$$F(y) = G(x) + C,$$

which usually defines y only implicitly as a function of x . Two practical points deserve emphasis.

KEY POINTS Handling $\int \frac{dy}{y}$

Whenever the integral produces a logarithm, write the absolute value: $\int \frac{dy}{y} = \ln |y| + c$. Do not drop the modulus signs. The sign of y is handled later, cleanly, by allowing the arbitrary constant to be positive or negative when you exponentiate: $|y| = e^c \cdot (\dots)$ becomes $y = K \cdot (\dots)$ with $K \in \mathbb{R} \setminus \{0\}$, and $K = 0$ is then checked separately.

WORKED EXAMPLE 3 Worked example 1: an implicit solution

Solve the differential equation $\frac{dy}{dx} = \frac{xy}{1+x^2}$.

SOLUTION

Step 1 — Constant solutions. Here $h(y) = y$, which vanishes at $y = 0$. The constant function $y = 0$ is indeed a solution (both sides equal 0).

Step 2 — Separate. For $y \neq 0$:

$$\frac{dy}{y} = \frac{x}{1+x^2} dx.$$

Step 3 — Integrate. The left side gives $\ln |y|$. For the right side, notice that the numerator x is (up to a factor 2) the derivative of $1+x^2$:

$$\int \frac{x}{1+x^2} dx = \frac{1}{2} \ln(1+x^2) + c.$$

Hence

$$\ln |y| = \frac{1}{2} \ln(1+x^2) + C.$$

This is the implicit solution. (Note: $1+x^2 > 0$ always, so no modulus is needed there.)

Step 4 — Make y explicit. Exponentiating,

$$|y| = e^C (1+x^2)^{1/2}.$$

Writing $K = \pm e^C$ (any non-zero real), and observing that $K = 0$ restores the constant solution $y = 0$ lost in Step 2, the general solution is

$$y = K\sqrt{1+x^2}, \quad K \in \mathbb{R}.$$

Check. If $y = K\sqrt{1+x^2}$, then $\frac{dy}{dx} = \frac{Kx}{\sqrt{1+x^2}} = \frac{x}{1+x^2} \cdot K\sqrt{1+x^2} = \frac{xy}{1+x^2}$. ✓

3.4 From Implicit to Explicit Solutions

The relation $F(y) = G(x) + C$ is a perfectly acceptable answer — it is the **general solution in implicit form**. But whenever algebra permits, we solve it for y to obtain the **explicit solution** $y = \varphi(x)$. This is possible when F can be inverted: typical tools are exponentiation (to undo $\ln$), taking reciprocals, or taking roots.

PROPERTY Branches of an implicit solution

An implicit relation $F(y) = G(x) + C$ may define several branches $y = \varphi(x)$ (for instance, $y^2 = x + C$ gives $y = \sqrt{x + C}$ and $y = -\sqrt{x + C}$). Each branch must be checked against the original differential equation: only branches that satisfy it, on an interval where they are differentiable, are genuine solutions. (This verification is examinable.)

WARNING Extraneous solutions from algebraic manipulation

Operations performed after integration can smuggle in false solutions. Squaring both sides turns $A = B$ into $A^2 = B^2$, which also admits $A = -B$; taking square roots introduces a $\pm$ that must be analysed, not ignored. Rule of thumb: after any non-reversible manipulation, substitute your final formula back into the original equation and check the sign.

WORKED EXAMPLE 4 Worked example 2: an explicit solution

Find the general solution of $\frac{dy}{dx} = y^2 \sin x$.

SOLUTION

Step 1 — Constant solutions. $h(y) = y^2 = 0$ when $y = 0$. The function $y = 0$ satisfies the equation: it is a (singular) solution.

Step 2 — Separate. For $y \neq 0$:

$$\frac{dy}{y^2} = \sin x \, dx.$$

Step 3 — Integrate.

$$\int y^{-2} dy = \int \sin x \, dx \implies -\frac{1}{y} = -\cos x + C.$$

Step 4 — Make y explicit. Multiply by -1 : $\frac{1}{y} = \cos x - C$. Renaming the arbitrary constant as $c = -C$ (still an arbitrary real constant),

$$\frac{1}{y} = \cos x + c \implies \boxed{y = \frac{1}{\cos x + c}}.$$

Full answer. The general solution is $y = \frac{1}{\cos x + c}$ ($c \in \mathbb{R}$), together with the singular solution $y = 0$, which is not obtained for any value of c .

Check. If $y = (\cos x + c)^{-1}$, then $\frac{dy}{dx} = -(\cos x + c)^{-2}(-\sin x) = y^2 \sin x$. ✓

3.5 Domain Considerations

A solution of a differential equation is, strictly speaking, a function defined on an interval. The formula $y = \frac{1}{\cos x + c}$ from Worked Example 2 is meaningless where $\cos x + c = 0$: the solution curve blows up there. Hence:

KEY POINTS Maximal interval of a solution

- The solution must be continuous and differentiable on its interval of definition; it cannot jump across a vertical asymptote.
- The interval must lie inside the domain where the original equation makes sense (where g is continuous and $h(y) \neq 0$ along the solution).
- The **maximal solution** is the solution defined on the largest possible interval containing the initial point. For $y = \frac{1}{\cos x + c}$ with $c = 2$, the denominator never vanishes (since $\cos x \geq -1 > -2$), so the maximal interval is the whole of $\mathbb{R}$; with $c = 0$, the denominator vanishes at $x = \frac{\pi}{2} + k\pi$, and each maximal solution lives on a single interval $(-\frac{\pi}{2} + k\pi, \frac{\pi}{2} + k\pi)$.

This is why, at GCE Advanced Level, an answer such as “ $y = \frac{1}{\cos x + c}$ ” should ideally be accompanied by a remark on where it is valid — especially once an initial condition (next section) picks out one particular branch and one particular interval.

DID YOU KNOW? Why “separation” is everywhere

Separable equations are the workhorse of applied mathematics. Radioactive decay $\frac{dN}{dt} = -\lambda N$, Newton’s law of cooling $\frac{dT}{dt} = -k(T - T_{\text{env}})$, and the growth of a savings account at continuous interest $\frac{dM}{dt} = rM$ are all separable — and all solved by exactly the routine of this section. Master these four steps now, and Section 5’s applications will feel like revision.

EXERCISE 2 Practice

Solve the following equations by separation of variables, giving the general solution and stating any constant solutions:

1. $\frac{dy}{dx} = \frac{x}{y}$ (leave the answer in implicit form);
2. $\frac{dy}{dx} = y \cos x$, and state the maximal interval of the solution through $(0, 2)$;
3. $\frac{dy}{dx} = \frac{y^2}{x}$ for $x > 0$.

SOLUTION Exercises 1–3

1. Separate: $y dy = x dx$. Integrate: $\frac{y^2}{2} = \frac{x^2}{2} + C$, i.e. $y^2 - x^2 = c$ (hyperbolas). No constant solution since $h(y) = \frac{1}{y}$ never vanishes — but note $y = 0$ is excluded as the original equation is undefined there.
2. $y = 0$ is a constant solution. For $y \neq 0$: $\frac{dy}{y} = \cos x dx$, so $\ln |y| = \sin x + C$, giving $y = Ke^{\sin x}$, $K \in \mathbb{R}$ (including $K = 0$). Through $(0, 2)$: $K = 2$, so $y = 2e^{\sin x}$, defined and smooth on the whole of $\mathbb{R}$ — the maximal interval is $\mathbb{R}$.
3. $y = 0$ is a singular solution. For $y \neq 0$: $\frac{dy}{y^2} = \frac{dx}{x}$, so $-\frac{1}{y} = \ln x + C$, i.e. $y = -\frac{1}{\ln x + C}$, valid on intervals where $\ln x + C \neq 0$ within $x > 0$.

4**INTEGRATION TO OBTAIN GENERAL SOLUTIONS**

In the previous section we learned how to separate the variables of a first order differential equation: we transformed an equation of the form $\frac{dy}{dx} = g(x)h(y)$ into the form

$$\frac{1}{h(y)} dy = g(x) dx,$$

where all the y 's live on the left and all the x 's live on the right. Separation, however, is only half of the journey. The equation still contains differentials; we have not yet found y as a function of x . The decisive step is now to **integrate both sides**. This section explains exactly how to do that, how to handle the constant of integration, how to simplify the result (especially when logarithms appear), and how to check that the answer is correct.

4.1 The Fundamental Step: Integrating Both Sides

DEFINITION General solution by direct integration

Once the variables of $\frac{dy}{dx} = g(x)h(y)$ (with $h(y) \neq 0$) have been separated, the **general solution** is obtained by integrating each side with respect to its own variable:

$$\int \frac{1}{h(y)} dy = \int g(x) dx + C,$$

where C is an arbitrary constant of integration.

Why is this legitimate? If two functions of x have the same derivative, they differ by a constant. Writing $H(y) = \int \frac{dy}{h(y)}$, the separated equation says $\frac{d}{dx}[H(y)] = g(x)$: indeed, by the chain rule,

$$\frac{d}{dx}[H(y)] = \frac{dH}{dy} \cdot \frac{dy}{dx} = \frac{1}{h(y)} \cdot g(x)h(y) = g(x).$$

Integrating both sides with respect to x gives $H(y) = \int g(x) dx + C$, which is exactly the formula in the definition.

METHOD Integrating after separation

1. Separate the variables completely: $\frac{1}{h(y)} dy = g(x) dx$.
2. Integrate the left side with respect to y and the right side with respect to x .
3. Add one single arbitrary constant C (it is enough to put it on one side, since two constants C_1, C_2 combine into $C = C_2 - C_1$).
4. Simplify: combine logarithms, exponentiate if useful, and absorb constants.
5. State the general solution, in implicit or explicit form as required, indicating the values of the variable for which it is valid.

WARNING Do not forget the constant

Omitting the constant of integration does not give the general solution — it gives only one particular solution. Since a first order equation has a one-parameter family of solutions, the general solution must contain exactly one arbitrary constant. In the GCE examination, a missing $+C$ costs marks every time.

4.2 Integration Techniques You Will Need

Separable equations reduce the problem to two ordinary integrals, so all the integration skills